

A Function-Pinned Testbed for the Commutator Critic: Implications of Homotopy-Distilled PC–GPT-2 for the RelaLeap V4 Programme, with Engineering-Plan Amendments and the C4' Stage Specification

Companion note to *The Commutator Critic* (v1.1) and to Mesto (2026),
Homotopy Distillation of a Transformer into a Predictive-Coding Network

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Abstract

Mesto’s technical report establishes that a pretrained GPT-2 (124M) can be re-expressed, near-losslessly and reproducibly, as an error-parameterised predictive-coding (PC) network by homotopy distillation: a continuous deformation from a backpropagation anchor ($T=1$) to a fully relaxed PC regime (terminal $\tau=6.4$), with student–teacher prompt KL $\sim 10^{-5}$ nats at *every* milestone, now extended to a completed 5×10^7 -token production run (worst milestone KL 3.1×10^{-5} , 35.7 A100-hours). This note works out what that result means for the commutator-critic programme (RelaLeap V4) and its supporting library (`comcrit`). Five implications are developed. (1) Because fidelity holds at every rung, the homotopy is a *function-preserving family*: every quantity the critic measures — propagator response, module-level commutators, confusion structure — can be measured as a function of the learning rule with the function pinned, which is precisely the controlled experiment the Causal-Continual-Learning (CCL) theorem needs, and it yields a pre-registerable prediction for Mesto’s planned continual-learning benchmark. (2) Mesto’s headline gradient-alignment result (per-block PC/BP cosine ≥ 0.998) is an $h=1$, feature-level statement; by the insufficient-statistics proposition and its executable sandbox demonstrations, it does not determine horizon- h counterfactual behaviour — the interesting object is the trajectory linear response, which is exactly what the exact- D tangent machinery measures. (3) The homotopy solves the cold-start problem for the critic’s equilibrium (IFT) branch by validation continuation from the already-validated $T=1$ case. (4) The measured error-mass rank frontier (top 5% of cells $\approx 90\%$ of squared update mass, concentrated in blocks 2–5, teacher-free) is an empirically supplied footprint: a pair-action sparsifier, a testable prediction about where gating effects live, and a reminder that in the PC form gating is architecturally native — there is no global backward pass to intercept. (5) The two programmes’ reproducibility methodologies are the same discipline independently arrived at, and each has one export the other needs: the strict-envelope threshold rule flows into the C-stage protocol; the 5σ effect-size precheck flows into Mesto’s E-line. The second half of the note converts these into engineering: amendments to the `comcrit` roadmap (a PC step adapter, the IFT branch pulled forward under homotopy-continuation validation, a frontier sparsifier behind an admission gate, settle-tangent cost levers), a pre-registration protocol binding CCL to the continual-learning benchmark, and a complete agent-executable C4' stage specification with frozen numerics and an A100 budget comparable to the production run itself.

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Scope note. This note takes Mesto’s results as reported (replicated fidelity, gradient-space homotopy, rank-frontier measurements, MORK companion) and the V4 programme as specified in *The Commutator Critic* v1.1 with its validated sandbox and the `comcrit` v0.1.1 skeleton. Nothing here re-litigates either; the subject is their composition.

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1 The two programmes, and the joint they now form

1.1 What Mesto’s report establishes

Three results, in his own careful framing. First, *expressivity at scale*: the PC parameterisation, with prediction errors ε_ℓ as the only free inference variables and weight updates local to each block, *holds* a modern 124M transformer — prompt KL to the teacher $\sim 10^{-5}$ nats at every milestone of the deformation, across five replicate runs and a completed 5×10^7 -token production run, evaluated teacher-free at $\varepsilon=0$. Second, the *gradient-space homotopy*: the per-block PC

weight update grows from the anchor identity η at $T=1$ to full backpropagation magnitude (ratio 0.98) while the minimum per-block cosine to the backpropagation gradient never leaves $[0.998, 1]$. Third, the *rank frontier*: at terminal τ the top 5% of the 4.72M error cells carry $\approx 90\%$ of squared update mass, the frontier sits in blocks 2–5 (66% of frontier cells; the last three blocks hold 0.2%), and the structure survives replacing the distillation objective with teacher-free cross-entropy (91.6% vs 90.0%).

Equally important is what the report explicitly does *not* claim: that PC trains from scratch at this scale, or that the PC form forgets less or transfers better. The continual-learning comparison — the reason to prefer PC at all — is designed but unrun, listed as the immediate next milestone.

1.2 What the critic programme supplies

The V4 programme, validated end-to-end at toy scale, is a measurement instrument for exactly the question Mesto has queued: given a modular learner mid-training, what is the *causal, horizon-h, paired counterfactual effect* of protecting a module from an incoming update — measured through an analytic backbone (exact injected perturbation of the full optimizer state, forward-mode linear response along the CRN baseline, exact cross-curvature synergy identity) rather than learned from summary features. Its theoretical spine is the CCL theorem: forgetting is governed by module-level gradient-field commutators, and the critic is the instrument that measures the realized commutator/confusion structure.

The joint is then plain to state. **Mesto has built the substrate on which the CCL-motivated questions about predictive coding become measurable rather than aspirational; the critic is the instrument that makes his forthcoming benchmark mechanistic rather than aggregate.** The rest of this note makes that composition precise and turns it into an engineering plan.

2 Implication 1: the homotopy is the controlled experiment the CCL theorem needs

The single most consequential property of the result is easy to pass over: fidelity holds at *every* rung, not merely at the endpoint. The homotopy is therefore a **function-preserving family of learners** $\{\mathcal{L}_\tau\}_{\tau \in [\eta, 6.4]}$: the input–output map is pinned (to $\text{KL} \sim 10^{-5}$) while the parameterisation, the inference dynamics, and hence the *learning rule* vary continuously.

Every quantity the commutator critic measures is a property of the learning rule, not of the function: the injected perturbation u , the update-map Jacobian and its products (the propagator), the per-module first-order conflict $\langle g_A, DP_m g_B \rangle$, the cross-curvature terms, the realized commutator matrix. In ordinary comparisons (PC-trained model vs BP-trained model) these quantities confound learning rule with function: the two models differ in what they compute, so differences in measured confusion structure cannot be attributed. Along Mesto’s homotopy the confound is removed by construction. Measuring the critic’s quantities at the anchor and at terminal τ on the *same* GPT-2 isolates the learning rule as cleanly as the field currently knows how to do.

This composes with the CCL theorem into a pre-registerable bridge to Mesto’s benchmark. Write $K_{mn}(\tau)$ for the realized module-level commutator/confusion matrix measured at homotopy parameter τ (modules = the 12 blocks; estimator = the critic backbone’s first-order and

cross-curvature terms, plus discrete paired measurements where wanted). CCL bounds forgetting by bilinear forms in K ; therefore:

Prediction 1 (CCL bridge, to be pre-registered before the E-line runs). *If the PC form exhibits a continual-learning advantage over backpropagation fine-tuning at matched compute and matched seeds, then the measured $K(\tau=6.4)$ differs from $K(\tau=\eta)$ in the direction the CCL bound requires (smaller realized cross-module commutator mass on the task pair used, or a better-conditioned confusion graph). Conversely, if $K(\tau)$ is τ -invariant within measurement noise, CCL predicts the benchmark finds no advantage attributable to the parameterisation.*

Both directions are informative, and the second is cheap insurance: it converts a null benchmark outcome — a live possibility, see §3 — from a disappointment into a theorem-consistent measurement with a mechanism attached. The critic’s distinctive contribution here may well be a *mechanistic null*.

3 Implication 2: cosine 0.998 is the $h=1$ shadow, and the sandbox already showed why that settles nothing

Mesto’s gradient-space figure is the natural headline: through the whole deformation the per-block cosine between PC and BP weight gradients never falls below 0.998. A tempting reading is that the PC form is dynamically equivalent to backpropagation, so the continual-learning comparison is foregone. The V4 analysis says precisely why that reading is unlicensed.

Instantaneous per-block gradient alignment is a *feature-level* statement — norms and cosines of current gradients. The insufficient-statistics proposition, made executable in the sandbox, demonstrated that such features determine the $h=1$ first-order counterfactual *exactly* (D6a: τ_1 coincides to 5×10^{-15} across feature-identical states) while leaving horizon- h behaviour undetermined (τ_{10} spans sign-flipping ranges across the same states), and the MLP stress test showed the propagator — the trajectory’s linear response, running through the optimizer’s moment recursions — is where horizon transport actually lives (frozen- D collapsing to Spearman $\sim 0.16/-0.05/0.20$ while exact- D held ≥ 0.997). Cosine 0.998 at each instant is compatible with materially different trajectory response, because the settle changes the update map’s Jacobian, interacts with Adam’s preconditioner, and imposes the rank-frontier sparsity on which directions transport.

So the live hypotheses after Mesto’s Figure 2 are exactly two: (a) $PC \approx BP$ in continual-learning behaviour *despite* the settle — the instantaneous agreement propagates; or (b) $PC \neq BP$ *because* of the settle — the agreement is an $h=1$ shadow and the propagators diverge. Only propagator-level measurement distinguishes them, and that is the critic’s native object: the exact- D tangent through settle-plus-update. This is worth stating to Mesto directly: his Figure 2 is the $h=1$ shadow of the object his E-line cares about; the machinery to measure the $h \gg 1$ object exists and is validated.

Prediction 2. *If the E-line finds a PC/BP difference in forgetting at matched compute, the difference will be visible in the measured horizon- h propagator response and commutator terms at the corresponding τ , and will not be predictable from instantaneous per-block gradient geometry (norms/cosines) — i.e. the black-box feature ceiling of sandbox D6b will reproduce on this substrate.*

4 Implication 3: validation continuation for the equilibrium branch

The critic’s CAROM/PC specialization has always had a cold-start problem: the fixed-point (IFT) backbone — equilibrium sensitivities $\partial z^*/\partial\theta = -(\partial F/\partial z)^{-1}\partial F/\partial\theta$ obtained by CG solves — would be validated against nudged finite differences on a substrate where nothing else was yet trusted. Mesto’s homotopy dissolves this. His settle-then-local-update is a pure step function of $(\theta, m, v; \text{batch}, t)$: T steps of ε -descent with weights frozen (asserted at runtime), then a local AdamW step. At $T=1$ this step *is* the standard transformer update up to the anchor identity, where the exact- D tangent backbone is already validated (sandbox; C4). The validation plan is therefore a continuation: validate at $T=1$, ascend the rungs $T \in \{1, 2, 4, \dots\}$ checking backbone-vs-paired-rollout agreement at each, switch the tangent from unrolled-jvp to IFT/CG when T makes unrolling uneconomic, and require the two tangent modes to agree at the crossover rung. At no point is the instrument validated cold — which is Mesto’s own methodological principle (never leave a regime in which the object is a faithful copy of something trusted) applied to the instrument itself.

Cost accounting, honestly: exact- D through a T -step settle multiplies the per-step tangent cost by $O(T)$. Two levers exist, both already in the programme’s idiom. Tangent-column batching (comcrit Patch 1) amortises the linearisation across the action set. And Mesto’s rank frontier is a cost lever for the *critic*, not only for training: a frontier-compressed settle used inside the tangent propagation is a frozen- D -shaped approximation, and it is admitted the same way — probe-state agreement against the full-settle tangent (cosine ≥ 0.95 , magnitude ratio in $[0.5, 2]$), never by assumption.

5 Implication 4: the rank frontier as measured footprint, and gating as a native operation

The V4 transformer plan sparsifies the quadratic pair-action space with HDC footprint codes: encode each module’s do-footprint, propose pairs by approximate-nearest-neighbour search, measure only proposed edges. Mesto’s frontier measurement supplies an *empirically given* footprint on this substrate, for free: which cells (hence blocks) carry update mass, stable across τ at fixed depth, robust to removing the teacher. Three uses follow.

(a) A sparsifier with an admission gate. Propose pair actions and cross-curvature measurements on frontier-heavy block pairs (blocks 2–5 dominate). Like every surrogate in the programme, the frontier proposer is admitted by measurement: on a probe set, the top- k frontier pairs must recover $\geq 80\%$ of the total measured cross-curvature mass that exhaustive pairing finds, or the HDC path is used instead. The two proposers can also be run head-to-head cheaply at 12 modules (66 pairs — exhaustive is feasible at this scale, which is exactly why the admission statistics gathered here transfer credibly to GPT-2-medium and beyond).

(b) A registered prediction. The frontier should be *where the causal action is*:

Prediction 3. *Modules (blocks) carrying larger frontier mass exhibit larger median $|\tau_h^A|$ under single-module gating at matched α and h , with rank agreement (Spearman over the 12 blocks) ≥ 0.6 . Failure of this prediction is itself a finding: it would mean settle-energy attribution and causal update attribution dissociate at 124M, which bears directly on using the frontier for settle compression.*

(c) **Gating is architecturally native in PC.** In the BP form, the critic’s gate intercepts a global backward pass. In the PC form there is no global pass to intercept: the weight update of block ℓ is local in ε_ℓ , so gating is attenuating ε_ℓ (or masking frontier cells) before a local update — an in-place, sparse, incremental operation, which is also exactly the write pattern the MORK trie maintains well (his companion result). The V4 corollary “policy as commutator suppression” — the critic-driven router as an online enforcement mechanism for the CCL hypotheses — becomes mechanically *cheaper* in the PC form than in the BP form. This is a genuine point in PC’s favour that Mesto’s report does not claim and the critic can quantify: even at zero continual-learning advantage from the parameterisation alone, the PC form may admit cheaper commutator suppression, which is an advantage of a different and measurable kind.

6 Implication 5: methodology flows both ways

Mesto’s §4 lesson — thresholds frozen from a single pilot sat inside run-to-run noise; recalibrate to the strict envelope over $n \geq 3$ replicates under a rule fixed in advance; judge configuration comparisons at matched seeds on per-seed deltas — is the V-series fail-closed discipline independently rediscovered, down to the preference for rank and scale-invariant statistics. Two exports, one in each direction:

Into the C-stage protocol: adopt the strict-envelope rule explicitly. Where a C-stage gate threshold was frozen after C1 from limited replication, the protocol now states: thresholds touching run-to-run variability are set as the envelope over $n \geq 3$ replicates, under a rule fixed before the final replicate is seen. (The gates this touches: the probe-agreement acceptance bands and the validity-radius ρ/e thresholds as applied per-seed.)

Into Mesto’s E-line: the 5σ effect-size precheck with measured noise floors. Forgetting deltas between PC and BP fine-tuning at matched compute are exactly the kind of signal that can sit below replicate noise — the V2 tiny-utility regime, which the sandbox reproduced live and the precheck caught. Before burning the comparison grid, the E-line should measure its noise floor (≥ 64 CRN replicate pairs at ≥ 20 probe configurations) and verify the planted domain shifts produce forgetting differences $\geq 5\sigma$ between the arms’ *controls*; if not, strengthen the shift (bounded iterations, recorded), never the thresholds.

7 Engineering-plan amendments

7.1 comcrit roadmap changes

1. **New: PCStepAdapter (v0.2).** A step-function wrapper for settle-then-local-update learners: `settle(state, batch, T, eta_eps)` composed with the local weight step, pure end to end, with the weight-frozen-during-settle assertion imported from Mesto’s runtime checks. Two tangent modes: `unrolled` (jvp through the T -step settle; exact- D default) and `ift` (CG solve on the settle fixed point; promoted from v0.3, see next). The adapter is substrate-agnostic — FabricPC graphs and the causal-fibres sidecar satisfy the same protocol.
2. **IFT branch pulled forward (v0.3 $\rightarrow$ v0.2.5), gated by continuation.** The equilibrium tangent is admitted only where it agrees with the unrolled tangent at the crossover rung (cosine ≥ 0.95 , magnitude ratio $[0.5, 2]$, ≥ 30 probe states), per §4. CG numerics inherit the C5 spec: tol 10^{-6} , max 200 iterations, warm-started along the trajectory.

3. **New: FrontierSparsifier.** Pair-action proposal from an externally supplied error-mass ranking, behind the admission gate of §5(a); interface-compatible with the HDC proposer so the two are interchangeable and comparable. Frontier data enters as a plain ranked index list — comcrit does not compute settles it does not own.
4. **Settle-tangent cost lever.** A `compressed_settle` option on `PCStepAdapter` tangent propagation (top- k frontier cells only), admitted by probe agreement against the full-settle tangent; this is the critic-side twin of Mesto’s planned rank-frontier settle compression, and the admission statistics should be shared between the two uses.
5. **Confusion export gains a concrete consumer.** The `ConfusionGraph` dict export is extended with a writer targeting store-expression form (block-pair terms as sparse incremental writes), aligning with the MORK companion; Atomspace typed-link export remains the v0.4 item but is now specified against a real store layout rather than speculatively.
6. **Protocol text:** the strict-envelope threshold rule (§6) is added to the numerical-hygiene policy, which is binding on all stages.

7.2 Pre-registration protocol for the CCL bridge

To be frozen *before* Mesto’s continual-learning benchmark runs: (i) the estimator of $K_{mn}(\tau)$ (backbone first-order conflict on the diagonal blocks; measured cross-curvature on edges; both margin-qualified against measured floors); (ii) the τ rungs at which K is measured (anchor, one mid rung, terminal); (iii) the task pair (matching the E-line’s sequential domains); (iv) the statistic and direction of Predictions 1–3; (v) matched seeds and per-seed deltas throughout, per the shared methodology. The pre-registration is a one-page addendum, and its value is exactly its timing: after the benchmark runs, the bridge becomes storytelling.

8 Stage C4’: the critic on the homotopy (agent-executable specification)

C4’ complements C4 (the from-scratch toy transformer); it does not replace it. C4 validates the machinery where everything is cheap; C4’ deploys it on the first real, function-pinned substrate. Prerequisites: C0 gates green on the target install (`comcrit.selftest()`); `PCStepAdapter` unrolled mode passing its unit tests; Mesto’s checkpoints and settle code importable with the weight-frozen assertion intact. All V-series discipline applies: CRN pairing, exhaustive tables at train states, measured floors, effect-size precheck, per-seed reporting, frozen thresholds, one-line deviation log.

8.1 Substrate and rungs

- Model: the PC-GPT-2 (124M) student. Rungs: $\tau \in \{\eta (T=1 \text{ anchor}), \approx 1.0, 6.4 \text{ (terminal)}\}$, using the production-run checkpoints ($\eta = 0.10$ family). The anchor rung is the go/no-go: it must reproduce C4-class backbone behaviour (the step is the standard update there) before any settle-bearing rung is run.
- Modules: the 12 blocks ($M=12$); the head is a 13th module in the action table but excluded from pair proposals.

- Tasks: Task A = the OpenWebText shard’s held-out split (retention; frozen 200-batch eval set, seq len 512). Task B = a disjoint-domain shard (code or PubMed-style text; choose once, record); 400 adaptation updates at the production run’s optimiser settings, gradient clip 1.0. Same task pair to be used by the E-line, so the $K(\tau)$ measurements speak to Prediction 1 directly.

8.2 Decision states, actions, horizons

- Decision states every 5 adaptation updates: 80 per rung per seed; seeds: 3, matched across rungs (the homotopy’s function-pinning is only an isolation if the seeds are).
- Actions: ordinary; 12 single-block gates at $\alpha \in \{0.5, 0.1\}$ (24); head gate at $\alpha=0.1$; top-5 frontier-proposed block pairs at $\alpha=0.1$ (§5a), plus 2 control pairs from the frontier’s bottom quartile. Behaviour-policy mix and propensity logging as in V3.
- Horizons: $h \in \{1, 5, 10, 25\}$ weight updates (each update containing its settle).
- Gating locus, fixed now to avoid agent improvisation: the gate multiplies block ℓ ’s *local weight-update gradient* $\varepsilon_\ell \partial f_\ell / \partial W_\ell$ (equivalently, attenuates ε_ℓ in the weight step only) — it does not perturb the settle itself. Settle-side gating (masking frontier cells during inference) is a distinct action family, deferred to a C4’’ if C4’ passes.

8.3 Truth, backbone, floors

- Truth: CRN paired rollouts — both arms share the batch sequence, settle initialisations, and RNG; snapshot/restore of (θ, m, v) bit-exact (import the V3 restoration test; add Mesto’s weight-frozen settle assertion to it).
- Backbone: exact- D via `PCStepAdapter.unrolled` — jvp through settle and AdamW jointly; tangent-column batching mandatory (Patch 1). At the terminal rung additionally compute the `ift` tangent and the `compressed_settle` tangent (top-5% frontier) at 30 probe states each; admission bands: cosine ≥ 0.95 , magnitude ratio $[0.5, 2]$ against unrolled exact- D .
- Floors and precheck: $\sigma_{\text{noise}}(h)$ from 64 CRN replicate pairs at 30 probe states per rung (replicates resample the continuation batch sequence); mandatory 5σ effect-size precheck at (h^*, α^*) selection, bounded strengthening (increase domain-shift strength of Task B , ≤ 3 iterations, recorded).
- Curvature: true HVPs for cross-curvature at 124M are affordable at probe-state counts; empirical-Fisher block surrogates computed alongside at every state and reported with agreement statistics (this is the calibration a GPT-2-medium stage will rely on).

8.4 Gates

1. **Anchor go/no-go:** at $\tau=\eta$, backbone Spearman ≥ 0.4 vs paired truth at $h \leq 10$ among margin-qualified states, per seed (the C4 gate, verbatim). Failure halts C4’ — the substrate integration, not the science, is broken.
2. **Rung validity maps:** full (ρ_h, e_h) maps per rung; the settle-bearing rungs must show a nonempty validity region at $h \leq 5$; report how the radius moves with τ (this curve is

itself a result: it is the first measurement of how PC inference depth affects second-order predictability of intervention effects).

3. **Tangent-mode agreement:** `ift` and `compressed_settle` admitted per their bands; disagreement is reported, not patched.
4. **Prediction gates:** Prediction 3 evaluated as specified (Spearman ≥ 0.6 over blocks); Prediction 2’s feature-ceiling arm: a ridge/kNN regressor on per-block norm/cosine features must trail the backbone by the D6b pattern on this substrate — reported descriptively, pre-registered directionally.
5. **$K(\tau)$ deliverable:** the module-level confusion matrices at the three rungs, margin-qualified, per seed, exported via `ConfusionGraph` — the artefact Prediction 1 consumes when the E-line runs.

8.5 Budget, stop conditions, deliverables

Anchor rung: ≤ 8 A100-hours (no settle in the tangent; costs are C4-like at 124M with batched tangents and 80 states). Mid rung ($T \sim 10$): ≤ 12 A100-hours. Terminal rung ($T=64$ -class): ≤ 20 A100-hours with `compressed_settle` carrying the tangent load where admitted, else reduce action-table size per the frozen budget-cut rule (cut sweeps and pair actions, never seeds, probes, floors, or replicates). Programme ceiling for C4’: 45 A100-hours — comparable to the production run that built the substrate, which is the right order for an instrument deployment. Stop conditions: anchor gate failure; no valid cell at $h \leq 5$ on both settle rungs (report as a substantive negative about second-order theory under deep settles); ceiling reached. Deliverables: `RESULTS.md` in the V-series format, the three $K(\tau)$ exports, the validity-radius-vs- τ curve, tangent-mode agreement tables, and the pre-registration addendum (§6.2) — filed *before* the E-line’s first seed is opened.

9 Risks and honest uncertainties

(1) The gating locus in PC admits variants (weight-step vs settle-side); C4’ freezes one and defers the other — a reviewer may reasonably prefer the union, but improvising both under one budget is how protocols die. (2) Exact- D through deep settles may be costlier than estimated if jvp memory forces recomputation; the compressed-settle lever and the IFT crossover are the mitigations, both gated. (3) The function-pinning argument of §2 pins the *function*, not the loss landscape geometry around it; attribution of K differences to “the learning rule” means the parameterisation-plus-inference-dynamics package, and the note says so rather than implying finer resolution. (4) Prediction 3 may fail — settle-energy and causal attribution can dissociate; the spec treats that as a finding with direct consequences for frontier-based settle compression, not as a gate failure. (5) All of this leans on Mesto’s checkpoints and settle code being importable with their runtime assertions intact; the anchor go/no-go exists to catch integration rot before any settle-rung spend.

Closing

Mesto’s report ends by calling its two results halves of one programme — learning-side and inference-side of a store-native transformer. The composition drawn here adds the third half,

in the only sense in which a programme can have three: the *measurement* side. A function-pinned family of learners, an instrument validated from toy scale upward with its failure modes turned into regression tests, and a theorem that tells both what to predict — that is a complete experimental loop, and every piece of it now exists. The engineering content of this note is small precisely because the two programmes were already converging on the same discipline; what remains is to run C4' and file the pre-registration before the E-line makes it moot.

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